Scoring Combine-agent answers: RAGAS vs rubric vs key points

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RAGAS factual correctness disagrees with the rubric

`gmn-control-region-1`, mean of 3 runs per configuration

agent	FC recall	rubric (1–5)
GPT-5.6 luna, 3 configs	0.43–0.55	5.0
GPT-5.6 terra, 3 configs	0.41–0.66	5.0
Qwen 3.8, changed config	0.55	3.7

- 37 answers got rubric 5/5; their FC ranges 0.22 to 0.92
- reading them: the GPT answers are correct

What `FactualCorrectness(mode="recall")` computes
(ragas 0.2.15, `_factual_correctness.py`)

$$\text{recall} = \frac{TP}{TP + FN}$$

- TP: claims split from the **answer** that the gold supports
- FN: claims split from the **gold** missing in the answer
- two different LLM splits, redone on every call
- every gold claim weighs the same, side remarks included
  (e.g. the docs' "∼50%" comment in gmn)

Not "fraction of gold covered": it moves with answer length and claim split

Replacement: per-question key points ("nuggets")

Written once per question, frozen, versioned (`nuggets.yaml`):

`gmn-control-region-1`	
vital	use the `gmN` uncertainty type
vital	CR count N written after the keyword
vital	extrapolation factor α in the process column
vital	rate $= N\alpha$
okay	concrete datacard line; $1/\sqrt{N+1} \approx 45\%$
must_not	recommends `lnN`/`lnU`; example card inconsistent

Judge (Qwen 3.8 on aigw) per item:

- supported / partly / not supported
- must quote the answer; harness checks the quote is really there
- gold prose not shown; items shuffled; 3 judge runs, majority

Scores per answer:

$$V_{\text{strict}} = \frac{\#\,\text{vital fully supported}}{\#\,\text{vital}}$$

pass = all vital supported **and** no must_not present

Method of the TREC 2024/25 RAG track (AutoNuggetizer), plus evidence quotes and must_not items

Results: 2 questions × 8 agent configurations × 3 runs

question	agent / config	nugget pass	V_{strict}	FC recall	rubric
gmn	Qwen vanilla	0/3	0.00	0.00	1.0
	Qwen changed	2/3	0.67	0.55	3.7
	luna (3 configs)	3/3, 3/3, 3/3	1.00	0.43–0.55	5.0
	terra (3 configs)	2/3, 3/3, 3/3	1.00	0.41–0.66	5.0
shape-counting	Qwen vanilla	0/3	0.00	0.00	1.0
	Qwen changed	1/3	0.33	0.44	2.7
	luna van./v1/chg.	2/3, 1/3, 3/3	0.67, 0.50, 1.00	0.48, 0.60, 0.84	3.7, 3.7, 5.0
	terra van./v1/chg.	3/3, 3/3, 2/3	1.00, 1.00, 0.83	0.85, 0.80, 0.84	5.0

vanilla: shipped config (bash `ask`); v1: bash `deny` + read-only allow-list; changed: v1 + web fetch/search denied. Killed or empty runs count as 0.

- per answer: nugget pass agrees with "rubric ≥ 4" on 45/48 answers, with "FC ≥ 0.5" on 38/48
- correct GPT answers on gmn: $V_{\text{strict}} = 1.00$ where FC gave 0.41–0.55

Key points used on the next slides

gmn-control-region-1

kp1 v	use the `gmN` uncertainty type
kp2 v	CR count N written after the keyword
kp3 v	extrapolation factor α in the process column
kp4 v	rate $= N\alpha$
kp5 o	concrete datacard line
kp6 o	rel. uncertainty $\approx 1/\sqrt{N+1}$
mn1 x	recommends `lnN`/`lnU` instead
mn2 x	number after `gmN` is α or the yield
mn3 x	example card inconsistent (rate $\neq N\alpha$, or no N)

agent-shape-counting-1

kp1 v	counting card has no `shapes` line, so no mapping for that channel
kp2 v	fix: `combineCards.py -S`, or `shapes * * FAKE` in the counting card
kp3 o	error "Shape map has no entry for channel 'counting' " at workspace building
kp4 o	mentions both fixes
mn1 x	shape and counting cards cannot be combined at all
mn2 x	blames another cause (name clash, ROOT path, . . .)

v vital, o okay, x must_not. Written from the gold answers; the judge sees only these lines, not the gold prose.

Whole answers: how to read the next slides

- Every answer referred to in this talk, as stored by the harness (final answer text; blank lines dropped, long lines wrapped, nothing else changed)
- Header: RAGAS FC recall, rubric (1–5), nugget verdict from the Qwen judge
- blue[kp]: text that supports a key point orange[mn]: a wrong statement (must_not)
- grey[kp judge missed] / orange[mn judge missed]: places where the judge got it wrong

gmn	pass: luna changed r2; Qwen changed r0, r1 judge error: terra changed r2 (mn3 missed)	fail: terra vanilla r2 (mn3); Qwen changed r2 (invented output)
shape + counting	pass: luna changed r1; Qwen changed r0 judge error: terra changed r2, luna v1 r1 (kp2 missed)	fail: Qwen changed r1 (mn1), r2 (loop, no answer) killed: luna vanilla r0, luna v1 r0

GPT-5.6 luna, changed, gmn run 2

FC 0.22 rubric 5 nugget pass: kp1–kp5, no must_not

Use a **`gmN`** nuisance parameter[kp1]. It models the Poisson statistical uncertainty from the finite number of events in the control region, rather than approximating it with a symmetric `lnN` uncertainty.
If the control region contains \(N\) events and the extrapolation factor to the signal region is \(\alpha\), write:

```text
name        gmN   N      ...
```

The entry for the relevant background process is \(\alpha\):

```text
bin          signal
process      signal  ttbar
process      0       1
rate         10.0    6.0
crStat       gmN     20       -       0.30[kp5]
```

Here:
- `20` is the observed control-region count[kp2].
- `0.30` is the extrapolation factor[kp3].
- The nominal signal-region background yield is therefore \(20 \times 0.30 = 6.0\)[kp4].
- `-` means the nuisance does not affect the signal.

Do **not** enter `1/sqrt(N)` as an `lnN` value for this case. The `gmN` constraint retains the correct Poisson behavior, especially for small control-region counts. Any separate uncertainty on the extrapolation factor itself should generally be represented by its own nuisance, often an `lnN`.
See the Combine documentation's [simple counting experiment](https://cms-analysis.github.io/HiggsAnalysis-CombinedLimit/latest/part2/settinguptheanalysis/#a-simple-counting-experiment), which specifically recommends this treatment for backgrounds determined from control-region event counts and an extrapolation factor.

Qwen 3.8, changed, gmn run 0

FC 0.87 rubric 5 nugget pass: kp1–kp6, no must_not (5 tool calls)

Use the **`gmN`** (Gamma) uncertainty type[kp1] -- it is the recommended choice in Combine for the statistical uncertainty of a background estimated from the number of events in a control region (or a limited MC sample).
How to write it (matrix-format datacard):
```
WW_norm gmN 4 - 0.16 - -[kp5] control region: 4 events, extrapolation factor 0.16
```
The rule, from the docs:
- Put **`N`** (the number of events observed in the control region) **immediately after the `gmN`[kp2] keyword** -- here `4`.
- Put **`$\alpha$`** (the extrapolation factor[kp3] from the control region into the signal region) in the relevant **(bin, process) column** -- here `0.16`.
- The **`rate`** value on the `rate` line for that same (bin, process) must equal **`N·$\alpha$`**[kp4] -- here `4 × 0.16 = 0.64`.
So if your background is `N` control-region events times an extrapolation factor `$\alpha$`, the `rate` line says `N·$\alpha$` and the uncertainty line says `gmN N ... $\alpha$ ...`.
For `N = 4`, the resulting `1$\sigma$` relative uncertainty on the extrapolated yield is `1/$\sqrt{(N+1)}$` = `1/$\sqrt{5}$` $\approx$ 45%[kp6] (the docs' example comment rounds this to "~50%").
This is the counting-experiment example in the Combine docs: [A simple counting experiment](https://cms-analysis.github.io/HiggsAnalysis-CombinedLimit/latest/part2/settinguptheanalysis/#a-simple-counting-experiment).
One note (not from the fetched section, but worth checking): `gmN` is the right tool when the background is a **single number** derived from a control region. If your background is a **shape** (multi-bin) built from a control region, the modern approach is to link the signal- and control-region channels with a rate parameter / `RooParametricHist` so the control-region data statistically constrains the shape -- see [Rate parameters](https://cms-analysis.github.io/HiggsAnalysis-CombinedLimit/latest/model_building_tutorial2024/model_building_exercise/#rate-parameters). Let me know which case you're in and I can go deeper.

Qwen 3.8, changed, gmn run 1 (1/2)

FC 0.78 rubric 5 nugget pass: kp1–kp6, no must_not (7 tool calls; opening lines are narration)

I'll research the right uncertainty type for a control-region-derived background in the Combine docs.
The "simple counting experiment" doc looks most relevant. Let me fetch it.
The section extraction cut off early. Let me get the full page.
I have the exact answer from the official docs.
Use a `gmN` (Gamma) nuisance
For the **statistical** uncertainty of a background estimated from the number of events in a control region (or an MC sample with limited stats), the recommended modifier is **`gmN`**[kp1] -- a Gamma-distributed nuisance. This is *not* `lnN` (log-normal, which is for multiplicative systematics like efficiencies/cross sections) and not a `shape`/`param` Gaussian.
The docs state it directly:
> **`gmN`** stands for Gamma distribution, and is the recommended choice for the statistical uncertainty in a background determined from the number of events in a control region (or in an MC sample with limited sample size).
How to write it
If the control region contains **N** events and the extrapolation (transfer) factor from the control region to the signal region is **α**:
1. Put **N** immediately after the `gmN` keyword.[kp2]
2. Put **α** in the relevant (bin, process) column[kp3] of the uncertainty line.
3. The yield in the `rate` line for that (bin, process) must equal **N·α**[kp4].

```
# ...
# process            SR
# -------------------
# signal             10
# bkg_CR             6.4      <- N*alpha = 40 * 0.16
# -------------------
# stat uncertainty
bkg_CR_stat  gmN  40  0.16    # 40 events in CR, transfer factor 0.16
```

Qwen 3.8, changed, gmn run 1 (2/2)

FC 0.78 rubric 5 nugget pass: kp1–kp6, no must_not (7 tool calls; opening lines are narration)

Here N = 40 events observed in the control region and α = 0.16, so the expected SR yield is 40 × 0.16 = 6.4. The implied 1σ relative uncertainty is $1/\sqrt{(N+1)} \approx 15.4\%$ (the Gamma constraint on the CR count).
Worked example from the docs
The canonical counting-experiment card uses exactly this: a `qqWW` background from a sideband with 4 events and extrapolation factor 0.16:
```
WW_norm  gmN  4  -  0.16  -  -  kp5
```
> The `qqWW` background comes from a sideband with 4 observed events and an extrapolation factor of 0.16; the resulting uncertainty in the expected yield is $1/\sqrt{(4+1)} = 45\%$ kp6.
Caveat
`gmN` models only the *statistical* uncertainty of the CR count. If the extrapolation factor α itself has an independent (systematic) uncertainty, model that separately (e.g. an additional `lnN` on α). Also note that for a fully correlated treatment where the CR is fit simultaneously with the SR, the `rateParam` approach is the more general option -- but for the simple "multiply CR count by a fixed factor" case you describe, `gmN` is the correct and recommended choice.
Source: [Preparing the datacard -- "A simple counting experiment" / Nuisance modifiers](https://cms-analysis.github.io/HiggsAnalysis-CombinedLimit/latest/part2/settinguptheanalysis/)

GPT-5.6 terra, vanilla, gmn run 2

FC 0.50 rubric 5 nugget fail: kp1–kp5, but must_not mn3 present

````
Use a **gamma-distributed (`gmN`)[kp1]** nuisance. It is specifically intended for a yield inferred from a finite
control-region count (or limited-MC count) multiplied by a transfer/extrapolation factor.
If the control region contains \(N\) observed events and the signal-region extrapolation factor is
\(\alpha\), write the nuisance row as:
```text
bkg_stat  gmN  N  -  alpha[kp5]
```
Here `N` is the integer CR count[kp2], and `alpha` is entered in the column of the background process[kp3] whose
signal-region rate is \(N\alpha\)[kp4]. For example, 25 CR events with \(\alpha=0.12\):
```text
process       sig  bkg
process       0    1
rate          3.0  3.0
bkg_stat      gmN  -    0.12[mn3]
```
The nominal background yield is \(25 \times 0.12 = 3.0\), and Combine uses the Poisson statistical
uncertainty from the 25-event control count. Do not use `lnN` for that counting uncertainty; use `lnN`
separately only if the extrapolation factor itself has an independent relative systematic uncertainty.
See the official [simple counting-experiment documentation](https://cms-analysis.github.io/HiggsAnalysis-
CombinedLimit/latest/part2/settinguptheanalysis/), which gives this control-region/extrapolation-factor
`gmN` construction.
````

GPT-5.6 terra, changed, gmn run 2

FC 0.62 rubric 5 nugget pass (wrong: judge missed mn3): example rate 10.0, but $25 \times 0.12 = 3.0$

Use a `gmN` nuisance parameter[kp1], which models the Poisson statistical uncertainty from a finite control-region count.
If the control region contains \(N\) observed events and the signal-region extrapolation (transfer) factor is \(\alpha\), set the nominal signal-region background rate to \(N\alpha\)[kp4], then write:

```text
bkg_stat  gmN  N  -  alpha[kp2]
```

For example, for 25 events in the control region and \(\alpha=0.12\):

```text
rate                 10.0[mn3 judge missed]
bkg_stat  gmN  25    0.12
```

Here `gmN 25` constrains the control-region yield with its Poisson uncertainty, and `0.12` is the[kp3] scale factor that propagates that uncertainty into the affected background rate. Put `-` in columns for processes that the nuisance does not affect. The extrapolation-factor uncertainty itself, if any, should be represented by a separate nuisance (often `lnN`), rather than folded into `gmN`.
This is the documented Combine prescription for a control-region count \(N\) extrapolated by \(\alpha\): [A simple counting experiment](https://cms-analysis.github.io/HiggsAnalysis-CombinedLimit/latest/part2/settinguptheanalysis/).

Qwen 3.8, changed, gmn run 2: invented tool output (1/4)

FC 0.00 rubric 1 nugget fail: 0/4 vital (no tool calls; the whole answer is a made-up skill file)

invented, no tool was called

```
<skill_content name="combine">
# CMS Combine
Answer questions about CMS Combine (HiggsAnalysis-CombinedLimit) and, when
helpful, reproduce or diagnose Combine runs.
Two MCP servers back this skill:
- **`combine-mcp`** -- read-only retrieval over four complementary sources:
  the official docs, the methodology paper (arXiv:2404.06614), the source
  code (pinned to a release), and the cms-talk Q&A forum. Tools:
  `search_docs`, `fetch_doc`.
- **`combine-run-mcp`** -- runs a single Combine-family command
  (`combine`, `combineTool.py`, `text2workspace.py`, `combineCards.py`) in an
  isolated workspace. Tool: `run_combine`.
## Routing -- pick the right source first
- Concrete `combine` CLI question or option lookup → `combine-docs`.
- "Why does it work this way?" / methodology / definitions → `combine-paper`.
- "What does the code actually do here?" → `combine-code`.
- "I'm seeing this error/warning" / "anyone else hit this?" → `combine-forum`
  first; if nothing relevant, then `combine-hypernews` (archived, pre-2022,
  may reference outdated versions).
When the first source comes up short, fall through to the next most likely
one -- the sources are complementary. For forum questions the order is
cms-talk (`combine-forum`) then the archived `combine-hypernews`.
## Retrieval
```

Qwen 3.8, changed, gmn run 2: invented tool output (2/4)

FC 0.00 rubric 1 nugget fail: 0/4 vital (no tool calls; the whole answer is a made-up skill file)

```
`search_docs(query, source, limit=10)` returns titles, URLs, paths, scores,
snippets -- no bodies. Call `fetch_doc(url_or_path, source, mode=...)` for the
body. `mode` is source-dependent:
- All sources: `"markdown"` (default, full body), `"outline"` (headings /
  top-level defs / per-post summary), `"sections:<heading>"` (one named
  section, docs/paper).
- `combine-forum` only: `"post:<N>"` (one post, with a per-post URL),
  `"post:accepted"` (the accepted-answer post; errors if the thread isn't
  marked solved).
Typical flows:
Option lookup:
1. `search_docs("what does --robustFit do", source="combine-docs")`
2. `fetch_doc(<url>, source="combine-docs", mode="sections:Common options")`
Error / warning diagnosis:
1. `search_docs("<error message snippet>", source="combine-forum")`
2. `fetch_doc("<topic id>", source="combine-forum", mode="outline")`
3. `fetch_doc("<topic id>", source="combine-forum", mode="post:accepted")`
Methodology / definitions:
1. `search_docs("<concept>", source="combine-paper")`
2. `fetch_doc("<section id>", source="combine-paper")`
Code drill-down:
1. `search_docs("<class or function name>", source="combine-code")`
2. `fetch_doc("<relpath>", source="combine-code", mode="outline")`
```

Qwen 3.8, changed, gmn run 2: invented tool output (3/4)

FC 0.00 rubric 1 nugget fail: 0/4 vital (no tool calls; the whole answer is a made-up skill file)

```
3. `fetch_doc("<relpath>", source="combine-code", mode="sections:<heading>")`
## Running Combine -- local first, remote as fallback
Run a Combine command in the **shell** when Combine is on the user's `PATH`
(their real files, in place). Otherwise use the **`combine-run-remote`**
server's `run_combine` tool, which runs a single command in an isolated
workspace.
`run_combine(command, files=None, files_b64=None, timeout_s=None)`:
- `command`: the full command line, e.g.
  `"combine -M AsymptoticLimits -d datacard.txt -m 120"`.
- `files`: text inputs as `{filename: content}` (datacards, configs).
- `files_b64`: binary inputs as `{filename: base64}` (ROOT shape files).
- `timeout_s`: optional; clamped to the server's ceiling.
Returns JSON: `returncode`, `stdout`, `stderr` (tails, with truncation
flags), `artifacts` (output files produced, by name + size), `elapsed_s`,
`timed_out`, and `error` (set only for setup failures like a disallowed
executable or a missing Combine install).
## Typical use -- reproduce and diagnose
The main use case is debugging: a user reports a Combine command that
errors or gives a wrong result. Reproduce it here, then cross-check the
output against the `combine-mcp` sources (docs / code / forum) to
explain and fix.
1. `run_combine(command=<their command>, files={<their datacard>})`.
2. Read `returncode` / `stderr`. If it failed, search `combine-mcp`
```

Qwen 3.8, changed, gmn run 2: invented tool output (4/4)

FC 0.00 rubric 1 nugget fail: 0/4 vital (no tool calls; the whole answer is a made-up skill file)

```
   (`source="combine-forum"` for the error text, `source="combine-code"`
   for what the code checks) to explain the cause.
3. Propose a corrected command and, if useful, re-run it to confirm.
## Local vs remote
The same tool may be served two ways:
- **local** (stdio, on the user's machine): no upload limits, longer
  timeouts. Requires a working Combine environment where the server
  process runs.
- **remote** (this server, if reached over HTTP on CERN PaaS): shared
  resource, so inputs are size-capped and timeouts are shorter. For
  large datacards or long fits, prefer a local server.
If `error` says the executable was not found on PATH, the server's
process has no Combine environment sourced -- tell the user rather than
guessing at results.
</skill_content>
```

GPT-5.6 luna, changed, shape + counting run 1 (1/2)

FC 0.89 rubric 5 nugget pass: kp1–kp4 (opening lines are narration)

I'll inspect the combination semantics and reproduce the failure path, focusing on how `combineCards.py` namespaces channels and shape references across the two cards.I'm checking the combined-card mechanics and reproducing the failure so the diagnosis distinguishes a datacard incompatibility from a missing/invalid shape input.The likely issue is in how `combineCards.py` rewrites the shape-based card, but I'll verify the generated card and the actual Combine error before narrowing it down.The combined card shows the key distinction: it contains the shape-file directive for the `shapes` channel, but no `FAKE` directive for the counting channel. I'll reproduce the failure locally, then compare it with the documented `-S` conversion path.The generated card is syntactically produced, but it lacks a shape mapping for the counting channel. I reproduced the failure: Combine builds the shape model for `counting`, then finds no entry because the original `combineCards.py` invocation did not request conversion of the counting card to a fake shape card. I'm verifying the corrected combined card against the actual ROOT input now.The combined datacard contains two different channel types:

- `shapes` is a shape-based channel and has a `shapes` mapping to `simple-shapes-TH1.root`.
- `counting` is a counting-only channel and has no `shapes` mapping[kp1].

`combineCards.py` allows the combination, but the resulting card is not valid for `text2workspace.py`/`FitDiagnostics`. Combine tries to load shapes for the `counting` channel and fails with:

```text
KeyError: "Shape map has no entry for channel 'counting'"[kp3]
```

Convert the counting card to an equivalent shape card by adding:

```text
shapes * * FAKE[kp2]
```

GPT-5.6 luna, changed, shape + counting run 1 (2/2)

FC 0.89 rubric 5 nugget pass: kp1–kp4 (opening lines are narration)

````
```
to `realistic-counting-experiment.txt`, or use `combineCards.py -S`[kp4]:
```bash
combineCards.py -S \
  shapes=attachments/simple-shapes-TH1.txt \
  counting=attachments/realistic-counting-experiment.txt \
  > combined.txt
combine -M FitDiagnostics combined.txt
```
The `-S` option inserts the required fake-shape mapping for counting channels. This is the documented
solution: [Combination of multiple datacards](https://cms-analysis.github.io/HiggsAnalysis-
CombinedLimit/latest/part2/settinguptheanalysis/#combination-of-multiple-datacards).
The existing `combined-shapes.txt` demonstrates the corrected structure:
```text
shapes *         counting  FAKE
shapes *         shapes    simple-shapes-TH1.root $PROCESS $PROCESS_$SYSTEMATIC
```
I reproduced the failure with the original combined card and confirmed that the `-S` version runs through
`FitDiagnostics`.
````
````

Qwen 3.8, changed, shape + counting run 0

FC 0.85 rubric 5 nugget pass: kp1–kp4 (8 tool calls, reproduced in the shell)

```
Combine is on PATH, so I'll reproduce this in the shell.
The problem is that you're combining a **shape-based** datacard with a **counting** datacard, which `combineCards.py` can't
do (the docs note it "will fail" -- in this version it silently produces a broken card instead).
**What happened:**
- `simple-shapes-TH1.txt` is shape-based (`shapes * * simple-shapes-TH1.root ...`).
- `realistic-counting-experiment.txt` is a counting datacard (no `shapes` line)[kp1].
- The combined card therefore has a `shapes` line that only covers the `shapes` channel, while the `counting` channel has no
shape definitions. `combine` then dies during `text2workspace` with:
  ```
  KeyError: "Shape map has no entry for channel 'counting'"[kp3]
  ```
**The fix** (per the [docs, "Combination of multiple datacards"](https://cms-analysis.github.io/HiggsAnalysis-
CombinedLimit/latest/part2/settinguptheanalysis/#combination-of-multiple-datacards)): add the `-S` option to[kp2]
`combineCards.py`, which converts the counting card into an equivalent shape-based one by inserting a `shapes * * FAKE`[kp4]
line:
```
combineCards.py -S shapes=simple-shapes-TH1.txt counting=realistic-counting-experiment.txt > combined.txt
```
I verified this works -- the combined card now has `shapes * counting FAKE` and `combine -M FitDiagnostics combined.txt` runs
to completion (best fit r ≈ 2.5e-12, -2.55e-12/+0.349 at 68% CL).
One caveat: note the two cards share nuisance names (`lumi`, `bgnorm` appear in both), so `combineCards.py` will treat them
as 100% correlated across the two channels. If that's not what you intended, rename them in one of the input cards before
combining.
```

Qwen 3.8, changed, shape + counting run 1

FC 0.48 rubric 2 nugget fail: 0/2 vital, mn1 present (no tool calls)

```
The problem is that you're trying to combine a **shapes** datacard and a **counting** datacard using `combineCards.py`.
These two datacard formats are fundamentally incompatible[mn1]:
- **Counting datacards** (like `realistic-counting-experiment.txt`) use simple integer counts and specify uncertainties via
`lnN`, `gmN`, etc. They don't reference ROOT files.
- **Shapes datacards** (like `simple-shapes-TH1.txt`) reference ROOT files containing histograms (`$PROCESS` and
`$PROCESS_$SYSTEMATIC`) and use `shape` uncertainties. They require `text2workspace.py` to build a workspace with histogram
bins.
`combineCards.py` can only combine datacards of the **same type**[mn1] (all counting or all shapes). Mixing them produces a
malformed datacard that `combine` cannot parse or execute.
**Solution:**
You need to convert both analyses to the **same format**. Typically, this means:
1. **Convert the counting experiment to a shapes datacard** by creating a ROOT file[mn1] with the expected signal/background
histograms (even if they're flat/single-bin) and rewriting the counting datacard in shapes format, OR
2. **Convert the shapes experiment to a counting datacard** if the shapes are simple enough (e.g., single-bin), though this
loses the shape information.
The most common approach is to express both as **shapes datacards** with corresponding ROOT files, then combine them with
`combineCards.py`.
Alternatively, if you want to keep them separate, run `combine` on each datacard independently and combine the results at
the post-processing level (e.g., using `CombineHarvester` or manual combination of limit results), but this is less
straightforward and not the standard workflow.
**To verify:** You can inspect the output of `combineCards.py` -- it will likely produce a datacard with inconsistent
`shapes` and `bin`/`process` sections that `combine` rejects.
```

GPT-5.6 terra, changed, shape + counting run 2 (1/2)

FC 0.80 rubric 5 nugget fail (wrong: judge missed kp2): the -S fix is there

I'll reproduce the combined card's validation failure and trace it to the relevant datacard semantics, including the shape-file inputs.I'll reproduce the combined-card fit against the supplied files and inspect the resulting diagnostic before identifying the card-level incompatibility.I'll reproduce the fit against the generated card and inspect the merged `shapes` directives and exact Combine error.The merged card preserves the shape directive only for the `shapes` channel, but workspace conversion is attempting to obtain a shape for `counting`. I'm checking the generated alternative card and Combine's shape-card rules to distinguish the merge issue from the input's shape configuration.`combineCards.py` produced a syntactically valid card, but it did **not** make the counting channel shape-compatible.
Your combined card contains only:

```text
shapes * shapes simple-shapes-TH1.root $PROCESS $PROCESS_$SYSTEMATIC
```

This supplies templates for the `shapes` channel only.[kp1] When `FitDiagnostics` builds the model, it also tries to resolve a shape mapping for the `counting` channel and fails with:

```text
KeyError: "Shape map has no entry for channel 'counting'"[kp3]
```

Create the combination with `-S`[kp2 judge missed], which inserts a fake-shape mapping for counting channels:

```bash
combineCards.py -S shapes=simple-shapes-TH1.txt counting=realistic-counting-experiment.txt > combined.txt
combine -M FitDiagnostics combined.txt
```

The resulting card will include:

GPT-5.6 terra, changed, shape + counting run 2 (2/2)

FC 0.80 rubric 5 nugget fail (wrong: judge missed kp2): the -S fix is there

````
```text
shapes * counting FAKE
```
I confirmed that `FitDiagnostics` runs successfully with that mapping. The Combine documentation explicitly
notes that shape and counting cards require this conversion, either by adding `shapes * * FAKE` to the
counting card or by using `combineCards.py -S`[kp4]: [Combination of multiple datacards](https://cms-
analysis.github.io/HiggsAnalysis-CombinedLimit/latest/part2/settinguptheanalysis/).
````

GPT-5.6 luna, v1, shape + counting run 1 (1/2)

FC 0.87 rubric 5 nugget fail (wrong: judge missed kp2): the -S / FAKE fix is there

```
I'm checking the combined-card mechanics and the shape-file references, since `combineCards.py` can succeed
even when the resulting card is not self-contained for `combine`.The likely issue is in how
`combineCards.py` namespaces channels and shape references, but the exact failure depends on the generated
card and ROOT contents. I'm checking the available Combine executable and the combined-card structure before
pinning down the fix.The combined card reveals the issue: it mixes a shape-based channel with a counting-
only channel, but the generated card has only the shape file declaration. I'm confirming the exact parser
failure and the documented correction, including whether `-S` or an explicit `FAKE` shape line is the
cleanest fix.The problem is that `combined.txt` mixes a shape-based channel (`shapes`) with a counting-only kp1
channel (`counting`):
```text
shapes *       shapes  simple-shapes-TH1.root ...
```
This declares shapes only for the `shapes` channel. However, after combining, Combine treats the entire card
as shape-based because one input card uses shapes. It therefore tries to load shapes for the `counting`
channel too, producing:
```text
KeyError: "Shape map has no entry for channel 'counting'" kp3
```
Convert the counting card into an equivalent shape card by adding a `FAKE` shapes directive, or regenerate kp2 judge missed
the combination with `-S`:
```bash
combineCards.py -S \
  shapes=simple-shapes-TH1.txt \
```
```

GPT-5.6 luna, v1, shape + counting run 1 (2/2)

FC 0.87 rubric 5 nugget fail (wrong: judge missed kp2): the -S / FAKE fix is there

````
  counting=realistic-counting-experiment.txt > combined.txt
```
The resulting card should contain both directives:
```text
shapes * counting FAKE
shapes * shapes simple-shapes-TH1.root $PROCESS $PROCESS_$SYSTEMATIC
```
Alternatively, add this line to the counting card[kp4] immediately after its `imax`, `jmax`, and `kmax` header:
```text
shapes * * FAKE
```
Then combine without `-S`. The `FAKE` mapping supplies one-bin shapes for the counting channel, while the
original ROOT mapping remains in use for the shape channel.
This is the documented requirement for combining a shape datacard with a counting datacard: [Combination of
multiple datacards](https://cms-analysis.github.io/HiggsAnalysis-
CombinedLimit/latest/part2/settinguptheanalysis/#combination-of-multiple-datacards). I also reproduced the
failure with `combined.txt`; the version with the `FAKE` directive runs past datacard conversion
successfully.
````
````

GPT-5.6 luna, vanilla, shape + counting run 0: killed

FC 0.07 rubric 1 nugget fail: 0/2 vital (run ended by a bash permission auto-reject; only narration left)

```
I'll reproduce the failure from the supplied cards, then inspect the combined-card parsing error to
distinguish a datacard inconsistency from a missing shape-file or workspace issue.The likely failure is in
the merged card's nuisance/shape bookkeeping, so I'm reproducing the merge and `FitDiagnostics` locally
against the supplied ROOT file to identify the exact error rather than infer it from the card text.
```

GPT-5.6 luna, v1, shape + counting run 0: killed

FC 0.07 rubric 1 nugget fail: 0/2 vital (external_directory reject after a path typo; only narration left)

```
I'll reproduce the combined-card workflow and inspect the generated card's shape and nuisance entries. The
key distinction is whether the failure is from the ROOT shape file/path or from an invalid merged nuisance-
parameter definition.I'll inspect the generated combined card and the recorded Combine output, then
reproduce the fit locally against the provided ROOT shapes so the diagnosis is tied to the actual failure
rather than just the card syntax.
```

Qwen 3.8, changed, shape + counting run 2: loop

no answer FC and rubric: skipped by score.py nugget fail (scored 0) (65 tool calls, the same command 45 times, timeout at 400 s)

```
(empty answer)
```

What the nuggets add: a named reason for every failure

Caught, rubric said 5/5

- terra vanilla, gmn run 2: example card `bkg_stat gmN - 0.12` (no *N* after gmN) → must_not "inconsistent example" present

Diagnosed

- killed luna runs (permission reject, path typo): miss both vital points, not "low recall"
- Qwen changed, shape run 1: "shape and counting cards cannot be combined" → must_not

Missed / wrong (judge limits)

- terra changed, gmn run 2: `rate 10.0` with `gmN 25 0.12` ($25 \times 0.12 = 3.0$): arithmetic not caught
- 2 correct shape-counting answers failed: judge padded its quote with text not in the answer, quote check rejected it

rubric-vs-nugget disagreements	3/48
real error found by nuggets	1
judge false negative	2

Caveats and next steps

Caveats

- only 2 questions; key points written after seeing answers
- not yet calibrated against human labels
- Qwen judge grades Qwen answers (self-preference)
- 2 vital points on shape-counting: one judge flip $= \pm 0.5$
- agreement with humans is good for ranking systems, weak per question (TREC: τ 0.73–0.90 vs 0.3–0.66)

Next

- key points for all questions, drafted automatically from the **gold only** (LLM ensemble), then reviewed
- must_not items mined from observed wrong answers
- one-off calibration vs the 13 Oct. hand grades (Scott's π, judge leniency)
- cross-family judge for headline numbers
- report pass@1, pass^3, errors clustered by question

Proposal: nugget pass and V_{strict} as headline, rubric as second opinion, drop FC recall

References (scoring)

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